

Yumu Yang

yumuy2@illinois.edu

[iNSPIRE HEP](#) | [Google Scholar](#) | [GitHub](#) | [GitLab](#)

Room 451, Loomis Laboratory, Urbana, IL, 61801

EDUCATION

- **PhD in Physics** 2020 - Current
Advisor since 2022: Jorge Noronha University of Illinois Urbana-Champaign
- **BA (Hons.) in Physics and Psychology** 2016 - 2020
Dean's List, Psi Chi Member Colby College

RESEARCH INTERESTS

As a member of the MUSES Collaboration, I am interested in the novel equilibrium and out-of-equilibrium phenomena displayed by the hot and ultradense matter formed in neutron star mergers. These problems are beyond the reach of first-principle techniques, and in my research, I employ relativistic fluid dynamics to understand the dissipative properties of neutron star matter. In this era of multi-messenger astrophysics with data from LIGO/Virgo and astrophysical observations from NICER, my research provides theoretical predictions that can be compared with experimental observations to gain further insight into fundamental properties of matter in extreme conditions.

EXPERIENCE

- **Nuclear Theory – Noronha Group** 2022 - Current
Research/Teaching Assistant University of Illinois Urbana-Champaign
 - Developed and maintained the holographic module for the NSF MUSES Collaboration.
 - 2024 - 2025: Organized the MUSES Collaboration Seminars.
 - Determined the bulk viscosity of nuclear matter at saturation density; calculated new bulk-viscous transport coefficients in neutron star matter; studied the effects of the nuclear equation of state on the dissipative properties of neutron star mergers/inspirals.
- **Biological Physics – Golding Lab** 2020 - 2022
Research/Teaching Assistant – Golding Group University of Illinois Urbana-Champaign
 - Studied cell transcription with experimental techniques such as smFISH.
- **DarkSide Collaboration (dark matter experiment)** Summer 2019
Research Assistant Gran Sasso National Laboratory, Italy
 - Tested Silicon Photomultipliers in both room temperature and cryogenic conditions.
- **Colby College** 2017 - 2020
Research/Teaching Assistant
 - Studied autocatalytic reactions, modeled and simulated the perturbed Brusselator.
 - Built laser diodes.
 - Graded students' homework and tutored physics courses.

HONORS

- **Vijay R. Pandharipande Prize in Nuclear Physics** 2026
Competitive departmental research award (\$2000) University of Illinois Urbana-Champaign
- **Mavis Future Faculty Fellow** 2026-2027
Fellowship supporting PhD students preparing for academic careers (\$2000) University of Illinois Urbana-Champaign
- **Tau Beta Pi** 2022-Present
Engineering Honor Society University of Illinois Urbana-Champaign

TECHNICAL SKILLS

- **Programming Languages:** C++, Python, Matlab, Mathematica.

LIST OF PUBLICATIONS

C=CONFERENCE, J=JOURNAL, P=ARXIV PREPRINT

- [P.1] Johannes Jahan, Kevin P. Pala, **Yumu Yang** et al., "Studying the QCD Matter produced in Heavy-Ion Collisions using the MUSES Calculation Engine," [arXiv: 2606.26326 \[nucl-th\]](#).
- [P.2] Abhishek Hegade, **Yumu Yang** et al., "Dynamical tidal response of neutron stars as a probe of dense-matter properties," [arXiv: 2603.26886 \[gr-qc\]](#), submitted to PRD.
- [P.3] Musa Khan, Ayrton Nascimento, **Yumu Yang** et al., "Uncertainty quantification of holographic transport and energy loss for the hot and baryon-dense QGP," [arXiv: 2603.20482 \[nucl-th\]](#), submitted to PRD.
- [J.1] **Yumu Yang** et al., "Merging multidimensional equations of state of strongly interacting matter via a statistical mixture," *Phys.Rev.D* 113 (2026), [arXiv: 2601.07987 \[nucl-th\]](#).
- [J.2] Isabella Danhoni, **Yumu Yang**, Mauricio Hippert, and Jacquelyn Noronha-Hostler, "Symmetry Energy of 2+1-flavor dense quark matter from perturbative QCD," *Phys.Rev.C* 113 (2026), [arXiv: 2510.23984 \[nucl-th\]](#).
- [J.3] **Yumu Yang**, Nikolas Cruz Camacho, Mauricio Hippert, and Jacquelyn Noronha-Hostler, "Symmetry-Energy Expansion with Strange Dense Matter," *Phys.Rev.C* 113 (2026), [arXiv: 2504.18764 \[nucl-th\]](#).
- [J.4] **Yumu Yang**, Mauricio Hippert, Enrico Speranza, and Jorge Noronha, "Symmetry-energy dependence of the bulk viscosity of nuclear matter," *Phys. Rev. Lett.* 135 (2025), [arXiv: 2504.07805 \[nucl-th\]](#).
- [J.5] Mateus Pelicer et al., "Building Neutron Stars with the MUSES Calculation Engine," *Phys. Rev. D* 111 (2025), [arXiv: 2502.07902 \[nucl-th\]](#).
- [J.6] **Yumu Yang**, Mauricio Hippert, Enrico Speranza, and Jorge Noronha, "Far-from-equilibrium bulk-viscous transport coefficients in neutron star mergers," *Phys. Rev. C* 109 (2024), [arXiv: 2309.01864 \[nucl-th\]](#).
- [J.7] Robert Bluhm and **Yumu Yang**, "Gravity with Explicit Diffeomorphism Breaking," *Symmetry* 13 (2021), [arXiv: 2104.05879 \[gr-qc\]](#).
- [C.1] **Yumu Yang**, Mauricio Hippert, Enrico Speranza, and Jorge Noronha, "Bulk viscosity dependence on the symmetry energy," *Quark Matter* 2025.
- [C.2] **Yumu Yang**, Mauricio Hippert, Enrico Speranza, and Jorge Noronha, "Bulk viscosity transport coefficients in neutron star mergers," *Quark Matter* 2023, [arXiv: 2312.09364 \[nucl-th\]](#).
- [C.3] **Yumu Yang**, Mauricio Hippert, Enrico Speranza, and Jorge Noronha, "Second-Order Transport Coefficients in Neutron Star Mergers," *Hot Quarks* 2022, [arXiv: 2402.06878 \[nucl-th\]](#).

LIST OF ORAL PRESENTATIONS

- | | |
|---|---------------------------------------|
| • Strangeness in Quark Matter 2026 | 2026 |
| <i>Connecting heavy-ion collisions to strange dense matter in neutron stars</i> | University of California Los Angeles |
| • Workshop on High Baryon Density Physics in High-Energy | 2026 |
| <i>Nuclear matter EoS at high density region</i> | Lawrence Berkeley National Laboratory |
| • The MUSES Seminar Series | 2026 |
| <i>Merging multi-dimensional equations of state via a statistical mixture</i> | online |
| • Nuclear Physics in Mergers - Going Beyond the Equation of State | 2025 |
| <i>Dependence of the bulk viscosity of neutron star matter on the nuclear symmetry energy</i> | University of Washington |
| • Quark Matter 2025 (largest and most important conference in the field) | 2025 |
| <i>Dependence of the bulk viscosity of neutron star matter on the nuclear symmetry energy</i> | Goethe University Frankfurt |
| • Midwest Theory Get-Together 2024 | 2024 |
| <i>How does the neutron star bulk viscosity depend on the symmetry energy?</i> | Argonne National Laboratory |
| • Quark Matter 2023 | 2023 |
| <i>Far-from-equilibrium transport coefficients in neutron star mergers</i> | Hilton of the Americas, Houston |
| • Hot Quarks 2022 | 2022 |
| <i>Second-order transport coefficients in neutron star mergers</i> | Dao House, Estes Park, Colorado |
| • Midwest Theory Get-Together 2022 | 2022 |
| <i>Second-order transport coefficients in neutron star mergers</i> | Argonne National Laboratory |